Deal or No Deal AI

Project 3B — Multi-Agent Deal Review Pipeline

Weekly submission report — Draft
Reporting date: October 25, 2026, Pacific Time
Tagline: Every deal has a story. Our agents find the plot holes.
Scope: Local educational prototype using fictional commercial-loan documents.

1. Executive summary

Deal or No Deal AI is a multi-agent financial deal review assistant that turns commercial lending documents into structured, explainable, evidence-backed review findings. Its purpose is to help an analyst understand a deal and identify what still needs human attention.

Traditional credit reviews require analysts to reconcile financial statements, loan agreements, collateral evidence and policy requirements manually. A single AI-generated recommendation can hide calculation errors, unsupported assumptions or unresolved exceptions. We designed around a different principle: **AI should accelerate financial analysis without removing transparency or human accountability.**

Using LangGraph, the application coordinates extraction, financial, compliance and risk responsibilities. Nebius powers structured extraction and synthesis; Python calculates ratios, evaluates configured policies and assesses decision readiness.

Our demonstration centers on a fictional **USD1.2 million equipment loan**. Despite DSCR of **1.50x**, LTV of **60%** and an illustrative **Low** risk score, the system retains **Pending Review** because the annual-reporting covenant remains unresolved. This is the central design insight: **a financially attractive deal is not necessarily ready for approval.**

A live uploaded Standard Deal has completed the analysis and memo workflow, including browser source inspection and report download. The expanded general- upload path adds parallel specialist execution and human-review checkpoints; one complete fresh live walkthrough of that newest path remains the highest- priority verification task.

The project demonstrates purposeful specialization, reproducible calculations, structured handoffs and reviewable exceptions. It is an educational decision- support prototype using fictional deals and simulated roles.

2. Assignment alignment

Assignment requirement	Implemented responsibility	Evidence and boundary
Agent 1 extracts key terms	Extraction Agent	Live structured Nebius output with source handles; saved examples remain separately labeled
Agent 2 checks compliance	Compliance Agent	Deterministic configured sample-policy checks; no claim of regulatory certification
Agent 3 flags risks and summarizes	Risk and Summary Agent role	Python risk stage followed by live synthesis; distinct internal nodes

Assignment requirement	Implemented responsibility	Evidence and boundary
Orchestrator coordinates the pipeline	LangGraph Orchestrator Agent	State, dependencies, conditional routing, checkpoints and explicit continuation
Three or more specialist roles	Extraction, Compliance, Risk/Summary; financial support	Two normal LLM stages, not three independent LLM agents
Agent communication	Structured shared-state artifacts	Source-linked panel outputs and recorded handoffs; no fabricated agent chat
Sample review output	Source-linked Markdown credit memo	Actual completed uploaded Standard Deal evidence retained
Recording and repository links	Submission artifacts	**[NOT VERIFIED]** Links still need to be supplied

3. Business problem and opportunity

A commercial-loan review combines contracts, financial schedules, collateral information and policy requirements. A reviewer must reconcile reporting periods, calculate ratios, identify missing evidence and prepare a defensible explanation. Fragmented documents make those tasks repetitive and make inconsistencies harder to notice.

A single free-form model response can mix extraction, arithmetic and judgment without exposing which step failed. This prototype separates those responsibilities and makes intermediate results visible before a human records a decision.

The intended business opportunity is less document transcription and memo assembly, more consistent application of explicit rules, and earlier identification of unresolved issues. These benefits are proposed; analyst time savings and lending outcomes have not been measured.

4. Product vision and target users

The product moves the analyst from assembling a review to challenging a prepared, source-linked analysis. Human judgment remains necessary at both stages.

Before: manual review	After: assisted review
Read across documents and copy terms into a working file	Inspect structured terms beside their source references
Re-enter numbers and calculate ratios separately	Check reproducible Python calculations and their inputs
Compare figures with policy limits manually	Review explicit applicability, rule results and exceptions
Track missing information in disconnected notes	Inspect unresolved requirements and permitted next actions
Assemble a memo and record the decision separately	Review a generated draft, then record a separate human action

This demonstrates fewer manual transfers within the application. It does not establish a measured reduction in analyst time or eliminate evidence review.

User	Job to be done	Product support
Credit analyst / loan officer	Prepare a coherent deal review	Upload inspection, extracted terms, calculations and draft memo
Underwriter / Risk Manager	Challenge repayment assumptions and missing inputs	Source evidence, ratio definitions, risk drivers and evidence-review controls
Compliance Officer	Review applicable controls and policy findings	Explicit applicability, exceptions and policy-linked evidence
Credit officer / committee member	Understand the recommendation and decide what happens next	Readiness, unresolved issues, comments and role-controlled human actions

Roles are simulated. The application assists decisions; it does not originate, fund or authorize a real loan. Insurance and separate equity-investment reviews are possible future applications, not implemented MVP workflows. The investment perspective in this MVP concerns the same loan as a debt investment.

5. Product features and user experience

Screen	User objective and capabilities	Backend involvement and output
Deal Dashboard	Find deals needing attention	Scoped stored review states, readiness and human status; navigable deal summaries
New Deal	Start a sample or upload one fictional deal package	Parse/validate PDFs, show page evidence, prefill editable reporting context and explicitly start analysis
Review Workspace	Understand and challenge findings	Extracted terms, financial metrics, policy results, risks, evidence explorer and human-review controls
Agent Activity	See what executed and where work paused	Recorded stage events, specialist outputs, handoffs and failures; no invented execution history
Credit Memo	Read and export a consolidated review	Python-generated Markdown with findings, conditions, limitations and supporting references
Decision History	Review who did what and why	Separately recorded comments, status changes, timestamps and analysis snapshots

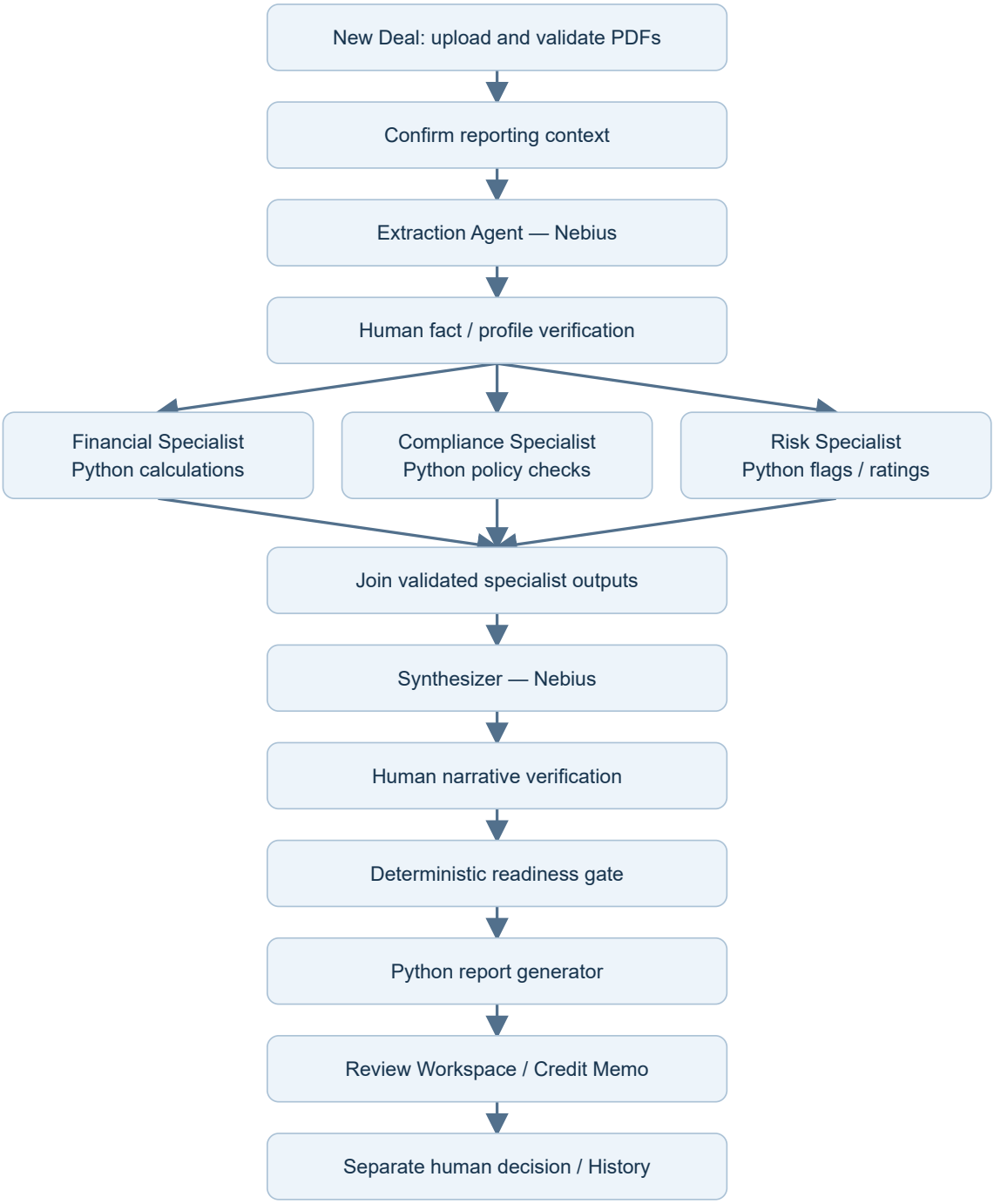
The reporting-context form uses document-derived suggestions where text is clear. It retains editable dates, annual actual/projection type and policy applicability; comparative dates are optional. Suggestions are not verification. Unrecognized or conflicting fields remain blank; policy applicability remains Unknown unless explicitly declared. Review date may default to the upload's UTC date, visibly identified as a default rather than a contractual date.

All downstream screens use the same scoped Deal ID, review identity, source binding and stored artifacts. Saving reporting metadata does not execute AI.

6. Multi-agent architecture diagram

Three mechanisms have distinct jobs: **LLM agents** interpret documents and explain findings; **deterministic specialist nodes** calculate and apply explicit rules; the **LangGraph orchestrator** manages dependencies, state and checkpoints. No additional planning LLM is needed to decide the prescribed next step.

The diagram below describes the **new uploaded-review path**. Registered sample and historical reviews retain their recorded sequential or demo-draft behavior.



LangGraph coordinates state, checkpoints and stage routing; SQLite retains records. Agent Activity reads recorded events. Diagram: new uploaded-review path.

7. Agent responsibilities and orchestration

Specialist independence and execution mechanisms

This is a **hybrid specialist workflow**: each specialist owns a bounded responsibility, consumes a defined contract and produces a separately validated output. Independence does not mean every specialist chooses its own goals or uses an LLM. Confirm this interpretation with the examiner if the rubric requires three autonomous LLM agents; that interpretation has not been verified.

Specialist	Independent responsibility	Execution mechanism	Output
Extraction	Identify deal terms and source evidence	Nebius LLM in `extractor`; Pydantic validation	Validated `DealTermsWire`, then scoped domain inputs
Financial	Calculate applicable metrics with consistent periods and units	Python specialist node `financial_analyst`	Financial results, validity states and calculation references
Compliance	Evaluate policy applicability and exceptions	Python specialist node `compliance_check`	Policy findings, unresolved applicability and exceptions
Risk	Aggregate risk drivers and available factors	Python specialist node `risk_quantifier`	Risk assessment, missing factors and risk flags
Synthesis	Combine three specialist results	Nebius LLM in `synthesizer`; structured-output checks	Structured recommendation, observations, rationale and conditions
Orchestrator	Coordinate routing, state and checkpoints	LangGraph outer graph and specialist subgraph	Recorded workflow state, artifacts and handoff events

Deterministic specialists suit arithmetic and explicit rules because identical valid inputs should produce identical results. Adding an autonomous model solely to increase the agent count would add cost and another source of inconsistency. The design demonstrates coordinated specialization; it does not claim autonomous reasoning at every node.

The report generator is a Python service, not an additional model call. Similarly, policy validation is part of the Compliance responsibility; it is not a duplicated policy LLM agent.

New uploads use a real three-branch LangGraph subgraph after fact verification. Branches return disjoint results and independently recompute inexpensive inputs. The parent joins and persists them, avoiding concurrent SQLite mutations. Synthesis does not start until the three panels are available.

Implemented graph execution and shared-state contracts

These node names reflect the implementation in `graph.py` and `live_worker.py`. This topology is **not** a claim that a fresh live execution has completed:

```
pdf_input -> extractor -> fact_review [human interrupt]
  -> specialist_panel
       |-> financial_analyst --|
       |-> compliance_check --|-> specialist_join
       |-> risk_quantifier ---|
  -> synthesizer -> evidence_review [human interrupt]
  -> route_review -> report_generator
```

Conditional routing may stop for missing information, invalid output, configuration or recovery, or add credit-review steps. A stop is not a completed stage. Registered samples and historical reviews retain their recorded topology.

Contract	Implementation and proof to show
Shared identity	`context.scope`: workspace, deal, review, run, thread and input revision
Model handoff	Validated extraction retains `value`, `refs` and `basis_id`; synthesis consumes current panels
Parallel fan-out	`run_specialist_branches` connects START to three specialist nodes; maximum concurrency three
Disjoint outputs	`PanelState.panels` merges branch-specific results; duplicate specialist output is rejected
Fan-in	`specialist_join` waits for all three branches; the parent persists their artifacts before synthesis
Recorded execution	Current-run events, artifact digests, execution-group and input-digest fields identify actual outputs
Human checkpoints	Fact/profile and narrative support records precede explicit graph continuation

An offline three-party threading-barrier test establishes actual fan-out. The fresh live recording must also show its events and joined results. Event insertion order alone does not establish parallel timing or a measured latency benefit.

SQLite-backed LangGraph checkpoints support pause/resume. Human evidence review uses interrupts and explicit continuation. Missing facts, invalid structured outputs or unresolved provider outcomes stop progression without substituting saved answers. Local checkpoints and audit records are retained, but are not claimed to be immutable or tamper-proof storage.

8. Critical design decisions and trade-offs

Problem and alternatives	Decision and rationale	Trade-off and mitigation
One large LLM prompt versus bounded specialists	Separate extraction, Python analysis, synthesis and routing so results can be inspected independently	More state/contracts to maintain; strict schemas and scoped artifacts limit error propagation
Model-generated ratios versus ordinary code	Calculate ratios and thresholds with Decimal	Less flexibility for unusual financial definitions; missing/unsupported bases are disclosed rather than guessed
Free-running supervisor versus a stateful graph	Use explicit LangGraph nodes, dependencies and human interrupts	Less autonomous planning; predictable progression suits the narrow demo
Sequential analysis versus parallel specialists	New uploads fan out Financial/Compliance/Risk and join; sample/historical behavior is preserved	Small repeated calculations; parent-only writes avoid state races. No measured speedup is claimed

Problem and alternatives	Decision and rationale	Trade-off and mitigation
Generate a memo immediately versus show intermediate review	Keep Workspace, source navigation and checkpoint reviews	More reviewer interactions; users can identify an incorrect period or unsupported claim before relying on the memo
Large retrieval infrastructure versus bounded document context	Use current deal's eligible cited pages directly for the MVP	Limited document size; benchmark retrieval separately before changing the runtime path
Treat AI Approve as final versus maintain separate decisions	Preserve AI proposal, deterministic readiness and human decision separately	More statuses; consistent labels explain why a positive proposal may still be Pending Review
Add many formats versus finish a narrow supported path	Keep the verified runtime intake to text PDFs	CSV, XLSX, DOCX and OCR remain unsupported in this intake until parser integration is validated

These choices are documented in the project specifications and implementation. The claim that specialist separation improves maintainability at larger scale is **[DESIGN RATIONALE — INFERRED]**, not a measured scalability result.

9. Implementation in brief

The application uses Python and Streamlit for the review experience, LangGraph for stateful coordination, Nebius for two normal generation stages, and Pydantic for structured contracts. Decimal calculations and policy rules remain in Python; SQLite retains scoped review records and checkpoints. pypdf supplies parsed text and page references.

Runtime grounding uses the current deal's bounded cited pages. It is not presented as a deployed vector-RAG system. Retrieval benchmarks and saved-answer evaluation are separate from the runtime review. The verified dependency inventory, code boundaries and retrieval limitations are collected in Appendix B.

10. Financial risk and compliance methodology

The calculations answer reviewer questions, not just populate a dashboard:

Finding	Meaning for the reviewer	What still needs judgment
DSCR1.50x	Supplied annual cash exceeds the documented annual debt service	Validate forecast assumptions and cash-flow evidence; full-term repayment remains unassessed
LTV60%	Supplied exposure is below the sample collateral threshold	Verify eligible value and lien premises; this is not a legal collateral opinion
Debt / EBITDA3.00x	Leverage is below the configured sample exception level	Assess the reliability of debt and earnings inputs
Projected revenue growth10%	The supplied forecast rises between the two periods	Recognize this as a projection comparison, not observed historical growth
Low illustrative risk	Available authored factors aggregate to a low modeled score	Do not interpret it as a default probability or policy clearance

Finding	Meaning for the reviewer	What still needs judgment
Covenant evaluation incomplete	Readiness remains Pending Review	Establish applicability, contractual timing, evidence and compliance status

The practical next action is to complete the unresolved review, not to approve because the numerical indicators look favorable. Definitions follow so another reviewer can reproduce and challenge each calculation.

The following definitions are implemented in Python. Inputs must refer to the same borrower, applicable period, currency and units. Definitions and thresholds are educational choices, not universal lending standards.

Metric / rule	Definition and required evidence	Standard Deal result / configured check
EBIT	Net income + interest expense + income-tax expense; annual schedule	USD900,000
EBITDA	EBIT + depreciation + amortization; no discretionary addbacks	USD1,100,000
Funded debt	Existing funded debt + proposed borrowing − documented repayment	USD3,300,000
DSCR	Annual cash available for debt service ÷ (annual principal + annual interest)	900,000 / 600,000 = 1.50x; sample exception below 1.25x
Debt / EBITDA	Funded debt ÷ annual EBITDA	3.00x; sample exception above 4.00x
LTV	(Secured exposure + prior-ranking liens) ÷ eligible equipment value × 100	60.00%; sample exception above 80% when equipment policy applies
Revenue growth	(Current revenue − comparative revenue) ÷ comparative revenue × 100	10.00%, comparing supplied projections
Current ratio	Current assets ÷ current liabilities	2.00x
Interest coverage	EBIT ÷ annual interest expense	9.00x
Debt-to-equity	Requires an approved equity-based definition and inputs	**Not implemented**; Debt / EBITDA must not be relabeled as debt-to-equity
Repayment capacity	Annual CFADS and documented annual debt service	Annual coverage only; balloon/refinancing/full-term adequacy not established
Covenant readiness	Applicability, timing, due date, evidence and compliance status	Annual financial-statement obligation remains incomplete without supported evaluation

Financial schedules supply the income statement, annual debt service and pro-forma balance sheet. Collateral pages supply value, exposure and lien premises. The loan agreement supplies terms and the annual-reporting covenant. Configured sample policy also includes term, amount and confidence-based review controls. Threshold breaches identify exceptions; they do not automatically mean Reject.

The illustrative composite score weights repayment capacity 30%, leverage 20%, collateral 20%, credit profile 20%, and business/industry 10%. It aggregates authored factor ratings, not a trained default model. The

Standard Deal's supplied ratings produce **21/100, Low**; this is not a 21% default probability. New generic uploads do not inherit these authored ratings and remain unscored when factors are unavailable.

Facts, arithmetic, policy findings, AI interpretations, missing evidence and human decisions remain distinguishable. Presence of a citation is not proof that its claim is supported. Future reporting obligations are not automatically breaches.

Educational approval exception: completed uploaded reviews with exclusively PENDING blockers may receive an explicitly acknowledged, reasoned human approval override from a simulated NORMAL Authorized Approver. The AI readiness remains Pending Review and the unresolved requirements remain recorded. Missing/stale evidence, material policy exceptions, recovery errors and outstanding information requests are not covered. This demonstration exception is not a real lending authorization or a model-generated waiver.

11. End-to-end demo walkthrough

Highest-priority proof: one fresh continuous walkthrough

Status: [NOT VERIFIED] for the newest general-upload live path. Historical live Standard Deal evidence and newer offline regressions establish different things. This report update adds a proof requirement; it does not execute a review.

Use a previously unused synthetic package with its own borrower and financial values. Retain one continuous recording; do not show saved Standard Deal results as if they came from the new upload.

Step	Required demonstration	Evidence to retain
1	Upload a previously unused synthetic deal package	Original package, filenames, file hashes and new Deal ID
2	Display parsing, reporting periods and extracted source evidence	Parsed pages, reporting declarations and cited preview
3	Trigger live Nebius extraction and inspect Pydantic output	Provider provenance, validated fields, source references and actual usage
4	Demonstrate human fact verification	Paused graph plus genuine reviewer assessments, including withheld support
5	Resume Financial, Compliance and Risk branches	Fan-out/join artifacts and stage events with one shared scope
6	Show merged findings and live synthesis	Three panels, structured recommendation, rationale and evidence
7	Complete narrative verification and readiness evaluation	Reviewer actions, continuation, unresolved requirements and readiness
8	Open Workspace, Activity, Memo and Decision History	Same Deal ID/review version, memo download and history without new inference

Retain an evidence manifest with Deal ID, run/thread identities, package digest, model and prompt/configuration versions, policy version, checkpoint/reviewer timestamps, provider usage and memo identity. A history page with

no human credit decision is valid; do not fabricate approval. If a decision is demonstrated, show its separate role, comments, timestamp and any educational-override label.

Completion requires evidence of all applicable stages and human checkpoints for that one review, with consistent results across screens. Report a partial run as partial, including its blocker. Unknown usage is not zero cost. A model or test helper must not manufacture human support for the actual demonstration.

Reviewer expectation: one continuous, reproducible execution is more persuasive than disconnected results from different versions.

Case: Example Equipment LLC, fictional USD1.2 million equipment term loan, 48-month term, supplied 2027 annual projections and 2026 comparative projections.

Demo step	User action	Agent / backend behavior	Evidence and result
Upload	Select the agreement, financials and collateral/assessment PDFs	Retain and parse one package	Original files, page text, validation and citation handles
Context	Review prefilled borrower, periods and applicability; confirm and save	Save an explicitly unverified reporting profile	Source suggestions and one upload-derived Deal ID
Start	Click Run once	Live Extraction Agent emits structured facts	Uploaded values and page references, not saved responses
Verify	Complete required checks in the indicated reviewer roles	LangGraph pauses/resumes for new upload evidence review	Recorded support or withholding, never automatic Pass
Analyze	Continue	Financial, Compliance and Risk branches join	DSCR, LTV, policy results, missing factors and risk flags
Synthesize	Continue after applicable checks	Live synthesis proposes a referenced explanation	Original AI proposal plus separate effective readiness
Review	Inspect Workspace and Agent Activity	Read the same deal's stored artifacts	Financial inputs, rule traces, source explorer and handoffs
Memo	Open Credit Memo and download	Python assembles upstream outputs	Findings, proposed conditions, limits and supporting sources
Decide	Authorized simulated reviewer records an action and comments	Separate version-checked human event	Decision History with UTC timestamp and retained analysis snapshot

The memorable result is **Low illustrative risk, zero numerical policy exceptions, but Pending Review**. The annual-statements covenant still needs its contractual timing and requirements evaluated. A Low score cannot complete that review.

Historical uploaded Standard Deal runs completed under the earlier unverified demo-draft policy. The new uploaded path adds mandatory fact and narrative checkpoints; do not present a historical draft as proof those newer human steps were performed. A missing-data case pauses or produces an explicitly partial review rather than inventing values or displaying a fabricated low score.

12. Evidence: what has been demonstrated

The evidence falls into four categories. Results from different checkpoints are not a single fresh full-suite run; overlapping test counts must not be added.

A. Actual live-model evidence

An uploaded Standard Deal completed a browser walkthrough with actual Nebius extraction and synthesis, source inspection, same-deal navigation and memo download. That review recorded **two generation requests**, **4,018 input tokens**, **1,109 output tokens**, and **USD0.0007345 generation cost at configured rates**. It remained an unverified educational draft with Pending Review readiness, not a credit approval.

This proves a bounded uploaded Standard Deal journey under its recorded policy. It does not prove completion of the newer general-profile human-checkpoint path. No fresh model call was made while revising this submission.

B. Deterministic and offline integration tests

Recorded test group	Observed result	What it establishes
Upload acceptance regression	57 tests and 55 subtests passed	Calculations, sample compatibility, citations, permissions and navigation under provider mocks
General-upload module	11 tests passed, including inherited cases	Different borrower/values, actual fan-out, checkpoint/resume, missing score and same-ID screens
Reporting-context save fix	Six focused tests passed	Metadata can save without clearing or depending on an unrelated execution lock
Educational approval override	Four focused tests passed; separate UI regression passed	Explicit authorized exception, unchanged readiness, role checks and visible controls
Recovery regressions	Ten tests passed	Scoped isolation retains unknown cost/history and fences replay

These results demonstrate software behavior; they are not model-accuracy scores. An independently read completed live record also contained the expected nine financial values and Pending Review readiness. References to the dated artifacts are in the evidence index.

C. Saved-answer evaluations and advisory judging

Assessment	Recorded result	Interpretation boundary
Native RAGAS on a saved Standard Deal answer	Faithfulness0.8; Response Relevancy approximately0.5363; Context Recall0.8	Not scores for a newly uploaded live answer
Native Context Precision	Not Applicable for that unranked DIRECT context	Do not invent a score
Separate custom five-dimension rubric	5/5 per dimension	Different scale and method; not interchangeable with native RAGAS

Assessment	Recorded result	Interpretation boundary
Fact-level judge	31 accepted citation checks; five flags among36 targets	Advisory citation assessment, not a validated extraction-accuracy percentage or human verification

The disagreement between custom perfect scores and lower native response relevancy deserves investigation. Different definitions, context, references or judge behavior may contribute; the cause is not established by those scores.

D. Future live-quality and business measurements

Not yet measured for the newest live path: field-level extraction quality, supported-claim rates, risk recall and false-positive counts, recommendation consistency, live-answer RAGAS, stage/end-to-end latency and repeat-run reliability. Phase 2 remains deferred until separately authorized. The proposed methodology and required references appear in Appendix D.

Analyst time savings, general accuracy percentages and parallel speedup are also unmeasured. A future user study should compare manual transcription, ratio preparation, policy lookup, evidence checking and memo preparation separately. Historical runtime troubleshooting is in Appendix C.

13. Business value and differentiation

The application reduces the need to transfer the same information between disconnected working files. Its intended contribution can be described precisely:

Manual task	Application contribution	Human work retained
Transcribe loan and financial terms	Produces source-linked structured fields	Check extraction, periods and document meaning
Recalculate financial ratios	Runs reproducible formulas on current inputs	Validate assumptions and economic interpretation
Look up sample policy limits	Displays predicates, observed values and outcomes	Determine applicability and resolve exceptions
Assemble a draft memo	Combines recorded findings and references	Review conclusions and unresolved issues
Reconstruct decision context later	Retains comments, timestamps and analysis snapshots	Own and explain the simulated decision

These are demonstrated workflow contributions, not measured productivity gains.

The demonstrated value is a connected review record: documents become cited facts, facts become reproducible calculations, calculations feed explicit rules, and findings become a reviewable memo and separate human action history.

Potential institutional benefits include reduced transcription, consistent application of approved rules, quicker draft preparation and clearer ownership of unresolved issues. These remain hypotheses for a future user study, not validated productivity or financial-return claims.

Differentiation is visible in concrete behavior: a two-call generation design avoids spending tokens on arithmetic; current-period/source scope prevents unrelated-deal leakage; partial information is exposed; and original AI

proposals are retained even when effective readiness disagrees. Those are useful product and engineering demonstrations without claiming enterprise readiness.

14. MVP limitations and prioritized roadmap

First priority: complete and record the newest general-upload workflow with genuine fact and narrative verification. This closes the largest submission- evidence gap before expanding formats, agent count or retrieval infrastructure.

Priority	Next work / limitation	Value, complexity and risk
P0	Finish one newly uploaded general-profile live walkthrough with genuine human evidence review	High demo value; moderate effort; closes the gap between offline generalization tests and live use
P0	Stabilize one documented app launch and improve provider diagnostics	High usability/reliability value; low-to-moderate effort; prevents confusing old-server and unknown-request states
P0	Reconcile historical uncertain provider usage when matching records are available	Cost transparency; external evidence dependency; no invented usage
P1	Phase 2 live-answer evaluation on held-out synthetic cases	High trust value; moderate effort; measure unsupported claims and excessive risk flags
P1	Test unfamiliar financial layouts, year-column alignment and cut-off passages	High error-reduction value; moderate effort; maintain explicit unsupported states
P1	Benchmark DIRECT, KEYWORD, VECTOR and HYBRID on identical questions	Determine retrieval value before adding complexity; do not assume vector search is better
P2	Additional formats, OCR and larger document packs	**[FUTURE ENHANCEMENT]** High breadth, higher parsing/context risk
P2	Real identity/authorization, institutional policy governance and protected audit infrastructure	**[FUTURE ENHANCEMENT]** Required before considering real financial-service deployment

Current intake is bounded to up to three synthetic text PDFs, ten pages and 10MB, USD whole units and annual analysis. Separate statistical default prediction, calibrated PD/LGD/EAD, real bureau/KYC integrations, portfolio concentration validation, multi-user load testing and production security are not established. The notebook artifact does not have a verified fresh Jupyter-kernel run; optional notebook installation was deferred. No public deployment is claimed.

15. Deliverables and links

Deliverable	Submission status
Application and specialized graph	Implemented locally; see architecture and limitations above
Sample memo and live evidence	Retained locally; select an approved synthetic artifact for submission

Deliverable	Submission status
Specification-driven documents	PRD, HLD, feature specs, execution plan, evaluation and traceability files
GitHub	**[INSERT GITHUB LINK — NOT VERIFIED]**
Demo recording	**[INSERT VIDEO LINK — NOT VERIFIED]**
Public application	**[NOT DEPLOYED]** Local instance is not a shareable hosting link

Suggested recording: show one package, inspect a source page, run the graph, explain a financial calculation, open its supporting evidence, demonstrate why Pending Review differs from Reject, and download the same deal's memo. Clearly label any educational human override and any saved response used in a fallback presentation; do not represent saved output as a new live call.

16. Conclusion

Deal or No Deal AI demonstrates a financial review workflow whose reasoning is inspectable at each handoff. Its strongest example is not an automatic approval: it is the ability to explain why apparently healthy financial metrics still leave an unresolved contractual review. The combination of bounded specialization, deterministic controls, source navigation and separately recorded human judgment makes the project a useful educational multi-agent implementation.

Evidence index

- Product requirements: PRD.
- Architecture: HLD, uploaded fan-out decision.
- Upload behavior: general upload contract, human decisions.
- Actual live/browser evidence: uploaded live results.
- Functional regression: acceptance fixes.
- Earlier evaluator evidence: answer-quality results.
- Runtime diagnosis: network context.
- Code: `graph.py`, `live_worker.py`, `finance.py`, `policy.py`, `risk.py`, `readiness.py`, `uploaded_decisions.py`, `memo.py` under `src/deal_or_no_deal_ai/`.

Some referenced documents retain historical status sections. Current code and explicitly dated execution evidence take precedence over an older status sentence.

Appendix A — Detailed specialist input and output contracts

Component	Receives	Produces and validates	Handoff
Extraction Agent	Current deal's parsed pages, source handles, reporting bases and strict schema	DealTermsWire: terms, financial inputs, evidence references and conflicts; Pydantic and source/basis checks	Human fact/profile verification

Component	Receives	Produces and validates	Handoff
Financial Specialist	Extracted values with units, periods and scope	Reproducible Decimal calculations; missing or invalid inputs remain explicit	Joined financial panel
Compliance Agent	Extracted facts, independently derived metrics and configured sample policy	Applicability-aware rule results, missing controls and exceptions	Joined compliance panel
Risk Agent	Current deal's facts, derived metrics and policy findings	Risk flags; weighted score only when supported authored factors exist	Joined risk panel
Synthesizer	The three recorded panels and allowed source references	Structured proposal, confidence, rationale, conditions and referenced observations	Human narrative verification and readiness routing
Orchestrator Agent	Shared state and explicit user commands	Ordered dependencies, fan-out/join where enabled, validation, budgets and conditional stops	Next permitted node or checkpoint
Credit Memo / Report Generator	Recorded upstream outputs and readiness	Deterministic Markdown report with evidence and audit references	Existing review screens and download

Appendix B — Detailed technology inventory and implementation boundaries

Verified technology	Role and fit	Limitation / trade-off
Python 3.12	Application, calculations, policies and report services	Local runtime and dependency setup required
Streamlit	Six-screen Python review interface	Rerun/session behavior needs careful identity handling
LangGraph	Stateful coordination, conditional routing and specialist subgraph	More explicit graph/state code than a linear script
langchain-core	Selected framework primitives	Does not imply every agent uses an autonomous LangChain executor
Nebius Token Factory	Live generation with Qwen/Qwen3-30B-A3B-Instruct-2507	Requires credentials, network access, token fit and available allowance
Pydantic	Strict extraction, synthesis and action contracts	Valid JSON does not prove factual correctness
pypdf	Local text-PDF parsing and page references	No OCR or guarantee of complex table/year-column fidelity
Decimal	Exact arithmetic and deterministic comparisons	Does not verify the economic meaning of an input
SQLite / LangGraph SQLite checkpointer	Scoped records, retained sources, checkpoints and action history	Local persistence; no enterprise access isolation or tamper-proof audit claim

Verified technology	Role and fit	Limitation / trade-off
pytest / Streamlit AppTest	Financial, graph, authorization and UI regressions	Mocked provider tests do not establish live model quality
Optional RAGAS 0.4.3 evaluation code	Separate answer-quality evaluation	Completed saved-answer evidence must not be presented as current live-answer evaluation

Core boundaries are implemented under `src/deal_or_no_deal_ai/`: upload intake and profiles; live context/worker; graph; finance/policy/risk services; readiness; uploaded decisions; memo; and shared UI projections. Prompts and specialist contracts are versioned under `docs/agents/`; model configuration is under `config/`. API credentials remain local and must not be committed or shown.

Retrieval clarification: runtime upload analysis uses bounded, cited document context. Parsing and placing pages in a prompt is not a vector RAG pipeline. Separate benchmark code implements DIRECT, keyword, vector and hybrid retrieval mechanics; hosted embedding capability has been checked. Those components do not establish a deployed hybrid retrieval path for every uploaded review. No Pinecone or other hosted vector database is required by the current workflow.

Appendix C — Historical runtime diagnostics

One operational lesson was directly observed: the earlier sandboxed runtime could not reach Nebius, while an approved read-only check outside it returned HTTP200 and confirmed the selected model. A network-capable local app was subsequently started. Credentials, network access and generation correctness are separate checks. Unknown request costs remain reserved rather than inferred from rounded aggregate bills.

This operational history explains reliability work; it is not the main demonstration or evidence of model accuracy. Local server addresses are session-specific, not public deployment links.

Appendix D — Proposed live-evaluation methodology

Status: proposed, not executed for this report. Freeze the final live review before evaluation; keep scores and test results separate.

Freeze the fresh review's actual extraction, context, calculations and synthesis before scoring. Establish expected facts independently from that package, checking borrower, year columns, units and qualifications. Do not reuse a saved reference answer for a different borrower.

Objective	Measurement on the frozen live review	Evidence boundary
Extraction	Correct expected fields, omissions and unsupported additions; counts and denominators	New live case not yet scored
Financial correctness	Independently expected Decimal results and missing/invalid-input behavior	Rule tests do not prove live extraction correctness
Policy correctness	Expected applicability, observed values and outcomes against the case-specific sample policy	No universal bank-compliance claim

Objective	Measurement on the frozen live review	Evidence boundary
Citation support	Source text supports the claim and correct period/borrower; count unsupported claims	Matching a source ID alone is insufficient
Risk identification	Expected risk recall and false-positive counts, reported separately	One demonstration is not broad accuracy evidence
Recommendation consistency	Readiness, unresolved requirements and next action agree	Original AI proposal remains distinct
Native RAGAS	Actual live answer/context/reference; metric version, raw score and status	Saved-answer scores are not live scores; unrun stays unrun
Custom rubric	Separate 1–5 scores and examples for faithfulness, relevance, completeness, attribution and coherence	Do not average or convert into native scores
Operations	Completion/stops, stage/total latency, tokens, known cost and unknown reservations	One run is descriptive, not a load benchmark

Investigate the existing custom 5/5 results alongside native response relevancy of approximately 0.5363. Different definitions, context, references and judge behavior may contribute; the cause is not established by the numbers. Similarly, 31 accepted citation checks from 36 fact targets are not a validated extraction accuracy percentage. Preserve disagreement and failures instead of selecting only favorable results.